2014 Tenth International Conference on Intelligent Information Hiding and Multimedia Signal Processing

Secure Speech Encryption System Using Segments for Speech Synthesis

Minoru KOHATA
Department of Information and Network Science
Chiba Institute of Technology
Narashino, Chiba, Japan
kohata@net.it-chiba.ac.jp

***Abstract*—A number of analog speech encryption methods have been proposed. At present, since most speech encryption systems use digital encryption, the demand for analog speech encryption is diminishing. However, the risk of speech tapping by decryption attack has increased as a result of the increasing availability of immense computing power and worldwide computer networks. In this paper, we propose a new secure analog speech encryption method that uses speech segments in a large-scale speech corpus as encrypted "transmission segments," instead of transmitting the synthesized speech itself. As such, the proposed method is robust with respect to decryption attacks because of the complexity of solving a permutation problem. Some experiments were carried out to evaluate the proposed encryption system, and the results show that it can transmit encrypted speech with good intelligibility, at a signal-to-noise ratio of 3 dB.**

***Keywords*-secure communications; speech encryption; speech coding; speech synthesis;**

I. Introduction

Several analog encryption methods were proposed before digital speech coding became common [1]. Most current speech encryption methods use strong digital encryption after using speech coding for information compression. As a result, the demand for classical analog speech encryption has diminished. However, due to the availability of enormous computing power and worldwide computer networks, the risk of a decryption attack has increased, but a combination of analog speech encryption and digital data encryption may be an efficient way to prevent this. In the present paper, we propose a novel speech encryption method that uses an analog speech waveform that carries the digital information required to synthesize the decrypted speech. By using a very large-scale speech corpus as the source of the analog speech waveforms, the proposed method is made robust against decryption attacks. The relevant background and a number of classical analog speech encryption methods are introduced in Section II. The proposed method is then described in Section III. A method for improving the performance of the proposed method is described in Section IV, and in Section V, we present the results of evaluating the performance through simulations and intelligibility tests.

II. Classical analog speech encryption

The advantage of analog speech encryption is that we can transmit encrypted speech over an ordinary voice transmission channel without the need for a codec. This form of encryption is carried out using various processes in the time or spectral domains, such as random replacement of data, inversion or transformation of the spectrum [2] [3], or insertion of dummy data between replaced or inverted data. [4]. The goal of analog speech encryption is to ensure that the encrypted speech is unintelligible and that the properties of the encrypted speech, such as the amplitude distribution or spectral bandwidth, are similar to those of a speech signal, because this makes it possible to transmit the encrypted speech over an ordinary speech transmission channel. However, it is not robust against decryption attacks, as described in [5], which reported that these classical methods can be decrypted by brute force, for example, by trying all possible permutations of frequency band replacements or inversions. Therefore, we propose an analog speech encryption method that is more robust against decryption attacks and can be used for practical secure communication.

III. Speech encryption with a large-scale speech corpus

Figure 1 shows a block diagram of the proposed method. Figure 1(a) shows the encryption step, and Fig. 1(b) shows the decryption step. The proposed method has two input modes: text and speech, and Fig. 1 shows the text input mode. In the text input mode, the speech corresponding to the input text is synthesized by a speech synthesizer, which contains the concatenated speech segments of the speech corpus. These segments are then scrambled using an encryption table. As a result, the original speech segments are replaced by segments from the speech corpus. The encrypted speech is then synthesized by concatenating the speech segments and truncating them to a fixed length (L). In the speech input mode, the speech segments that are close to the input speech are selected by using one-pass DP matching [6]. These segment numbers are then scrambled in the same manner as in the text input mode. We proposed this method, which is referred to as the coding by synthesis (CbS) method, in a previous study [7]. In this paper, we evaluate and discuss the text input mode only, because the performance of the speech input mode depends on the accuracy of the DP matching. In the decryption step, the encrypted speech is divided into segments of length L,

978-1-4799-5390-5/14 $31.00 © 2014 IEEE
DOI 10.1109/IIH-MSP.2014.72

IEEE computer society

the feature parameters are calculated for each segment, and the patterns of these parameters are then matched with the segments in the same corpus that was used in the encryption step. As a result, a series of the encrypted speech segments are obtained. We can decrypt these segments by using a table that provides information about how to replace the encrypted segments with segments of the original speech.

With this method, the following information is required for decryption:

1) a corpus for the transmission segments and a corpus for speech synthesis;
2) a pattern matching procedure and a defined feature parameter;
3) a decryption table.

Even if the first two of these items are disclosed and the number of transmitted segments are known, in order to decrypt the original segments by searching the all permutations, speech synthesis must be performed T times, to correctly decrypt the speech:

$$T(N_{seg}, M) = {}_{N_{seg}}\mathrm{P}_M, \tag{1}$$

where N_{seg} is the number of segments in the corpus, M is the number of transmitted segments, and ${}_{N_{seg}}\mathrm{P}_M$ denotes M permutations from N_{seg}. In general, N_{seg} is more than several ten of thousands and M becomes several hundred per sentence. Since as indicated by Eq. (1), the computational cost is factorial, a decryption attack performed by searching all permutations becomes impossible in practice.

IV. Selection of speech segments for encryption

In the proposed method, if the speech corpus contains segments that are similar (such as "silence" segments), it might be hard to discriminate between these segments in the decryption process. In addition, if the encrypted speech is contaminated with channel noise, the decoded speech quality will be degraded by "segment error". In order to avoid this, we should remove these segments from the speech corpus before encryption. If too many segments are thus removed from the corpus, it is possible that it can become too small to be sufficient for speech synthesis. For this reason, we prepared an initial corpus that is several times larger than that required for speech synthesis, then we designed the corpus for transmission segments in addition to the corpus for speech synthesis. The relationship between the corpus for speech synthesis and that used for the transmission segments is shown in Fig. 2. In order to optimally decimate the segments, we have to examine all combinations of the segments to be decimated. This has a high computational cost, so we propose using the following method to decimate the segments more efficiently.

1) For the large-scale corpus, we used the corpus used in the Japanese TTS system XIMERA [8]. The specifications of this corpus and the calculations of the feature parameters are shown in Table I. We used "ATR503 sub-corpus" for the speech synthesis, which is one of the sub-corpora of this corpus.
2) Truncate all the segments in the corpus so that they are all of length L. Segments for which the duration is less than L or for which the power is less than P_{th} are decimated, since these segments increase errors in the decryption process. After this decimation, DFT with a Hamming window is used to calculate the feature parameters of all of the truncated segments.
3) The distance between segments is defined by Eq. (2), where $X_i(n)$ denotes the n-th DFT coefficient of the i-th segment and N_{DFT} is the dimension of the feature

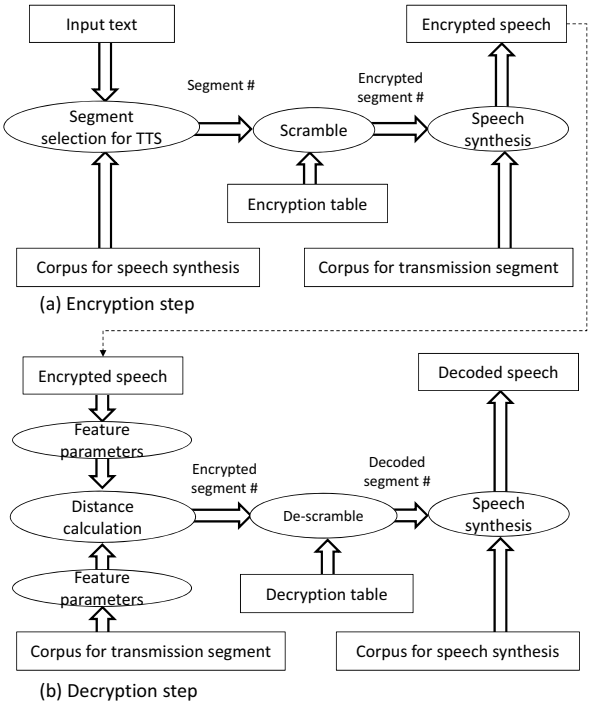


Figure 1. *Block diagram of the proposed speech encryption method.*

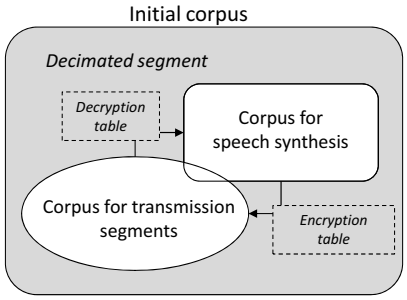


Figure 2. *Relationship between the corpus for speech synthesis and the corpus for transmission segments.*

Table I
SPECIFICATIONS OF SPEECH CORPUS.

Speech synthesizer	XIMERA
Sampling freq.	16 kHz
Initial corpus size	24,8012
Corpus size for TTS (ATR503 sub-corpus)	31,842
Speaker	female announcer
Segment duration (L)	40 ms
Power threshold (P_{th})	65 dB
Feature parameter	DFT (1024 point)
Dimension of feature parameter (N_{DFT})	200 (real:100 imaginary:100)

parameters:

$$d(i,j) = \sqrt{\sum_{n=0}^{N_{DFT}} (X_i(n) - X_j(n))^2 / N_{DFT}}. \quad (2)$$

4) If $d(i,j) < th$, then decimate the latter segment, that is, the j-th segment, as follows:

```
for(i=0; i<Nseg-1; i++){
  if(seg(i).alive == false) continue;
  for(j=i+1; j<Nseg; j++){
   if(d(i,j) < th) seg(j).alive = false;
  }
}
```

Here, N_{seg} is the number of segments in the initial corpus and $seg(j).alive$ is an indicator that is $true$ initially, but becomes $false$ if the j-th segment is decimated.

This method tends to decimate the segment whose index j is larger than i, then it is not an optimal method. However, it will be effective to reduce the error rate, since the distance between any two segments becomes larger than th after the decimation. Also, the computational cost of this method is less than $O(N_{seg}^2)$, so it is practical. Figure 3 shows the relationship between the threshold (th) for decimation and the decimation rate. The dotted line corresponds to the size of the corpus used for speech synthesis. The size of the corpus used for the transmission segment must be larger than that used for speech synthesis, so the decimation rate should be above this line.

V. EXPERIMENTS

A. Evaluation of the segment error rate

We carried out simulated experiments in order to evaluate the segment error rate of the proposed method under noisy conditions. In the experiment, we assumed the text input mode shown in Fig. 1, and the encryption table was made by randomly mapping a segment in the corpus for speech synthesis to one of the segments in the corpus for transmission. Speech synthesis was carried out for a very long text;

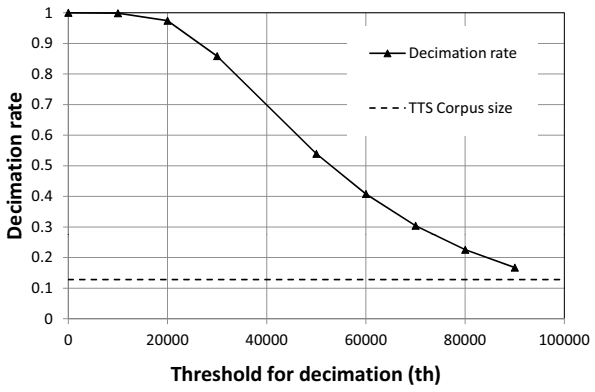


Figure 3. *Relationship between decimation rate and threshold for decimation.*

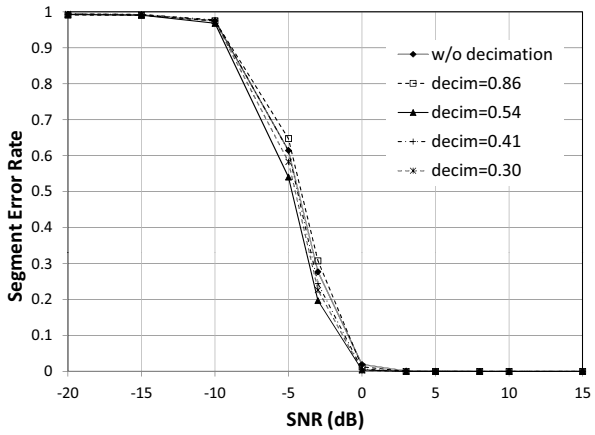


Figure 4. *Relationship between the SNR and the segment error rate.*

the selected segments were encrypted and then replaced by the transmission segments. The total number of segments for this test was 11,822. The transmission segments were then contaminated by white noise with a specified signal-to-noise ratio (SNR). White noise can never simulate the noise of a real speech-transmission channel, of course, but we wanted to first evaluate the proposed system under a canonical condition. Figure 4 shows the result of the segment error rate (SER) measured at the specified SNR. The SER remained at a very low rate under very noisy conditions, such as SNR = 0 dB. As shown in Fig. 4, a decimation rate of 0.54 resulted in the lowest SER. A lower decimation rate means that the transmission segments are sparsely aligned, and so the SER should decrease, if the decimation is optimal. However, the proposed decimation method is not optimal, and we assume that the SER could not be further reduced with a lower decimation rate.

B. Intelligibility test for decoded speech

Finally, we conducted an intelligibility test for the speech decoded from encrypted speech that had been transmitted

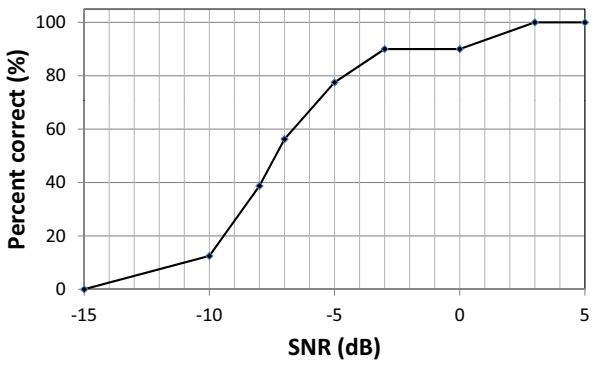


Figure 5. *Relationship between intelligibility and the SNR.*

through a noisy channel. We adopted an intelligibility test rather than a quality test, since the quality of the decoded speech was suddenly degraded as the SNR decreased, then, for example, a subjective quality test using mean opinion score (MOS) would be inefficient. In this experiment, the synthesized speech element was the name in Japanese of a single digit number (0 to 9). This was encrypted by the proposed system, and white noise was added to the encrypted speech as in the previous experiment. The encrypted speech was then decrypted, decoded (with segment error), and then synthesized. The decoded speech was then presented to subjects, who identified the digit (0 to 9, or unidentified) that they heard. The intelligibility of the decoded speech was defined to be the mean score of the correct answer. The test was carried out with seven Japanese male subjects. We used a decimation rate of 0.54, since it had been found to minimize the SER in the previous experiment. Figure 5 shows the results of the intelligibility test: all of the decoded speeches for which the SNR was greater than 3 dB were correctly recognized. Assuming that the encrypted signal will be coded by a speech coder such as μ-law or A-law, the SNR of the coded signal will be at least 3 dB. This means that the proposed encryption system has satisfied one of the conditions required for practical use over a real speech transmission channel. We think that the following is the reason why the encrypted speech is robust against noise. The transmission segment is created from actual spoken words, so its spectrum has a wide band structure, whether it is from a voiced period or from an unvoiced period. That is, the transmission segments have a spread spectrum, and this may make the system robust against noise, because spread spectrum signals usually have a high autocorrelation.

VI. Conclusion

We have proposed a speech encryption method that uses speech segments in a large-scale speech corpus. In the proposed system, instead of transmitting the original speech, an encrypted speech segment is used as a transmission segment. We presented a method for designing the corpus for the transmission segments and a convenient segment selection method was proposed; we determined that this method was effective for decreasing the segment error. The performance of this system was evaluated, and the result showed that the segment error rate (SER) of this system stayed low under noisy conditions. An intelligibility test was also conducted, and the decoded speech was properly recognized when the SNR was 3 dB. In order to make this system useful in practice, we have to overcome some problems other than channel noise, such as nonlinear effects, group delay effects, temporal skew effects, distortion caused by the codec, and the naturalness of encrypted speech. We will consider these in future work, and we will also continue developing the speech input mode.

Acknowledgment

This study was supported in part by a grant of the Strategic Research Foundation Grant-aided Project for Private Universities from the Ministry of Education, Culture, Sport, Science, and Technology, Japan (MEXT).

References

[1] N. S. Jayant, B. J. McDermott, S. W. Cristensen, and A. S. Quinn, “A Comparison of Four Methods for Analog Speech Privacy, “ IEEE Trans. on Commun., Vol. COM-29, No. 1, pp. 18–23, Jan. 1981.

[2] B. Goldburg, S. Sridharan, and E. Dawson, “Design and Cryptanalysis of Transform-Based Analog Speech Scramblers,” IEEE Journal of Selected Areas in Communications, Vol. 11, No. 5, pp. 735–744, June 1993.

[3] A. Jameel, M. Y. Siyal, and N. Ahmed, “FFT Based Analog Speech Scrambler using TMS320C6711 DSP,” IEEE 9th International Multitopic Conference, pp. 1–4, Dec. 2005.

[4] A. Matsunaga, K. Koga, and M. Ohkawa, “An Analog Speech Scrambling System Using the FFT Technique with High-Level Security,” IEEE Journal of Selected Areas in Communications, Vol. 7, No. 4, pp. 540–547, May 1989.

[5] B Goldburg, E Dawson and S Sridharan, “The Automated Cryptanalysis of Analog Speech Scramblers,” Advances in Cryptology EUROCRYPT 91, Lecture Notes in Computer Science, Vol. 547/1991, 422-430, 1991.

[6] L. Rabinar and B. Juang, “Fundamentals of Speech Recognition,” pp.416–420, Prentice Hall, 1993.

[7] M. Kohata, “A low bit rate speech coder based on concatenative speech synthesis”, 2009 Spring Meeting Acoustical Society of Japan, 3-6-12, pp.367–368, 2009.

[8] H. Kawai, T. Tomoki, J. Yamaguchi, T. Hirai, J. Ni, N. Nishizawa, M. Tsuzaki, K. Tokuda, “XIMERA : A Concatenative Speech Synthesis System with Large Scale Corpora”, IEICE. Trans. Information and Systems, Vol. J89-D-II, No. 12, pp.2688–2698, 2006.